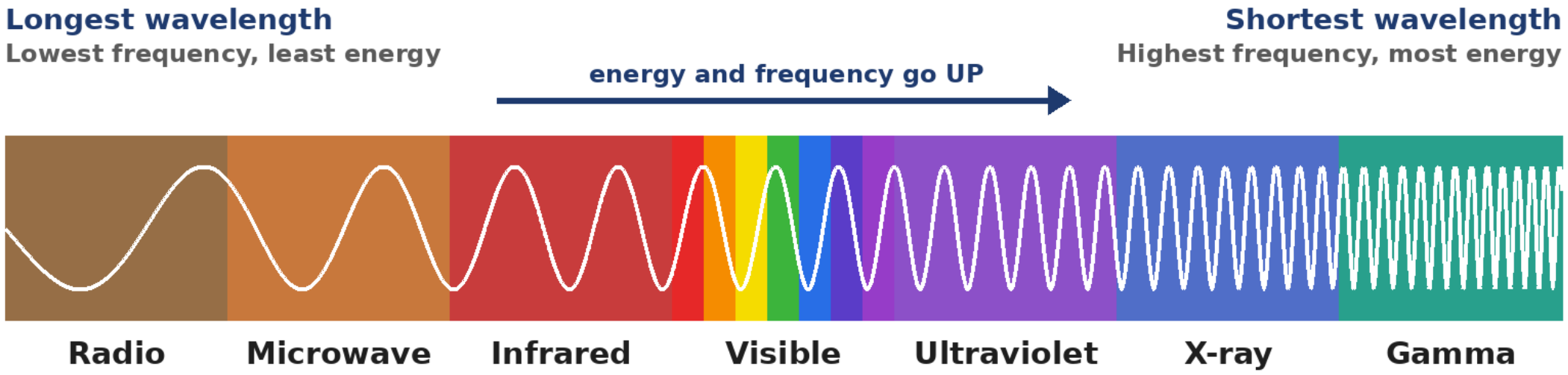
4.08 Electromagnetic Spectrum

What makes each kind of light wave different?



Visible light is the only part we can see. Red = longest, violet = shortest.

Watch First: The EM Spectrum



Today's Goals

I can put the EM spectrum in order from longest to shortest wavelength.

I can explain how frequency, wavelength, and energy are linked.

I can match each EM wave to how we use it.

I can use what I know to escape the EM escape room.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Light and other EM waves do not need a medium. They can travel through a vacuum (empty space).
- They can also travel through some materials, like clear glass.
- As the frequency of a wave goes up, its wavelength goes down.

In this lesson

- EM spectrum video
- Integers: The 4 Rules + Case #1
- Part 1: EM Spectrum drag and drop
- Part 2: EM Spectrum Escape Room

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

INTEGER CREW: THE 4 RULES

Adding + Subtracting Integers

1 • SAME SIGNS? TEAM UP!

Add the counts. Keep the sign.



$-6 + (-3) = -9$

2 • DIFFERENT SIGNS? FACE OFF!

Subtract the counts. Keep the sign of the greater absolute value.

Greater absolute value = farther from zero.



$-7 + 3 = -4$

3 • SUBTRACT A POSITIVE? FLIP IT!

Keep the first. Add the opposite.



$7 - 10 = 7 + (-10) = -3$

4 • SUBTRACT A NEGATIVE? FLIP IT!

Keep the first. Add the opposite.



$-9 - (-2) = -9 + 2 = -7$

Subtract? KEEP the first. CHANGE to +. FLIP the second!

Robots represent signed numbers.

CASE #1 • MONDAY

Does the storm alert stay on?

WAVE WORD: *Infrared*

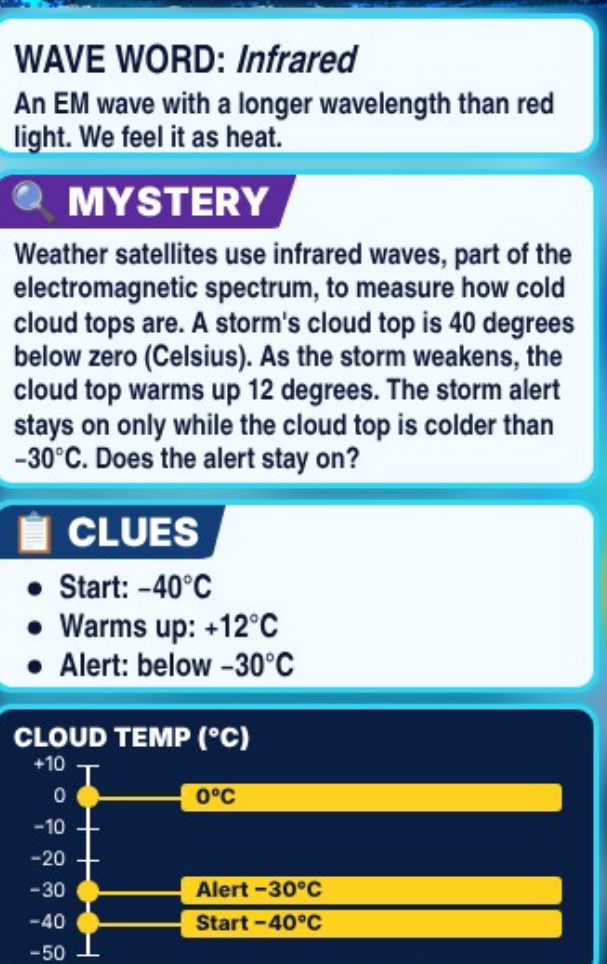
An EM wave with a longer wavelength than red light. We feel it as heat.

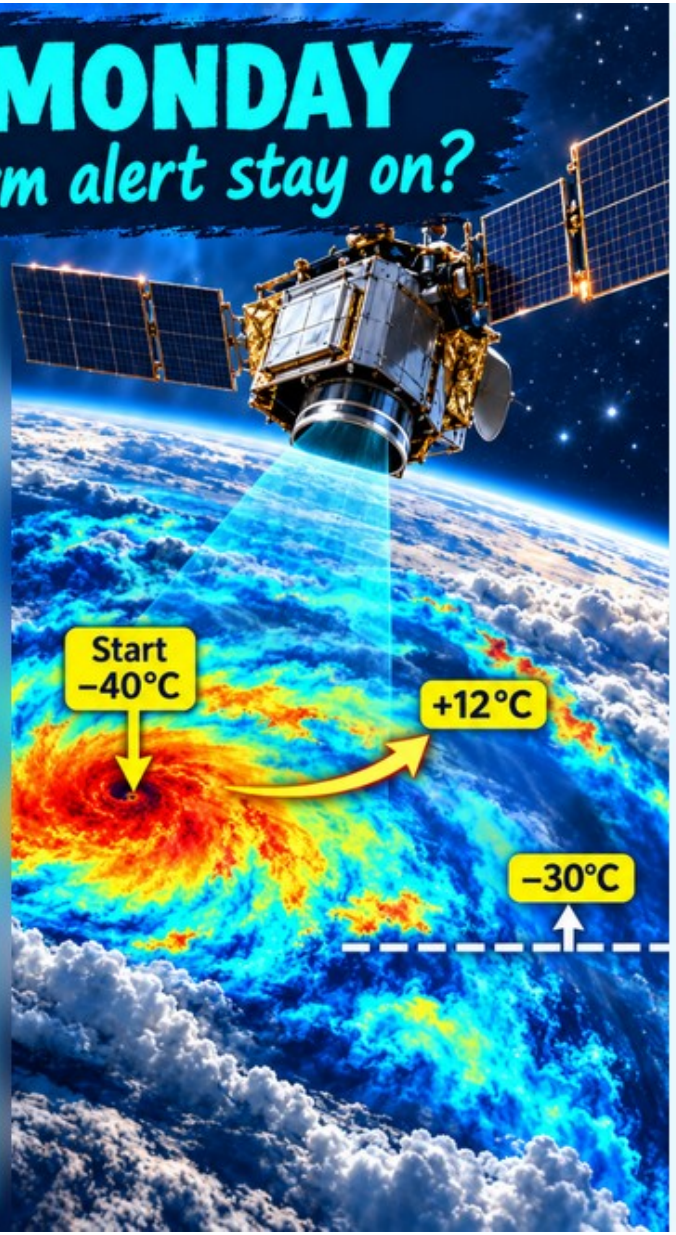
MYSTERY

Weather satellites use infrared waves, part of the electromagnetic spectrum, to measure how cold cloud tops are. A storm's cloud top is 40 degrees below zero (Celsius). As the storm weakens, the cloud top warms up 12 degrees. The storm alert stays on only while the cloud top is colder than −30°C. Does the alert stay on?

CLUES

- Start: −40°C
- Warms up: +12°C
- Alert: below −30°C





CLAIM

Does the storm alert stay on?

☐ YES ☐ NO

EVIDENCE

Solve:

−40 + 12 = _____

The cloud top is now _____ °C.

REASONING

Finish the sentence:

Different signs: I subtract 40 − 12 = _____ and keep the sign of _____.

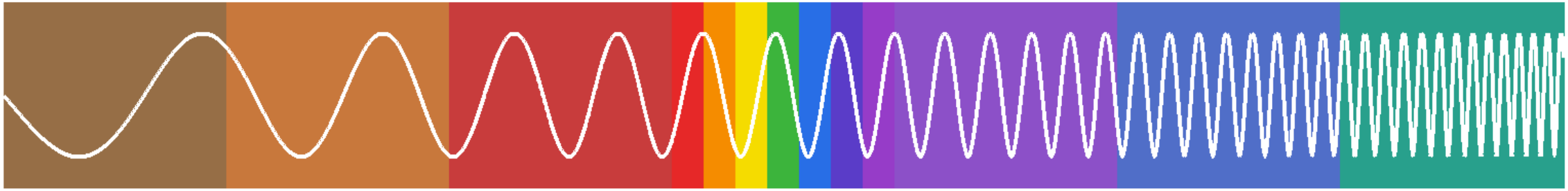
−28°C is _____ than −30°C.

Part 1: Put the Spectrum in Order

Longest wavelength
Lowest frequency, least energy

Shortest wavelength
Highest frequency, most energy

energy and frequency go UP



Radio Microwave Infrared Visible Ultraviolet X-ray Gamma

Visible light is the only part we can see. Red = longest, violet = shortest.

Remember the order: **Raging Martians Invaded Venus Using X-ray Guns**
All EM waves travel through space at the speed of light. Higher frequency = shorter wavelength = more energy.

Engageli Poll

Which EM wave has the MOST energy?

Single choice poll

A. radio waves

Single choice poll

B. visible light

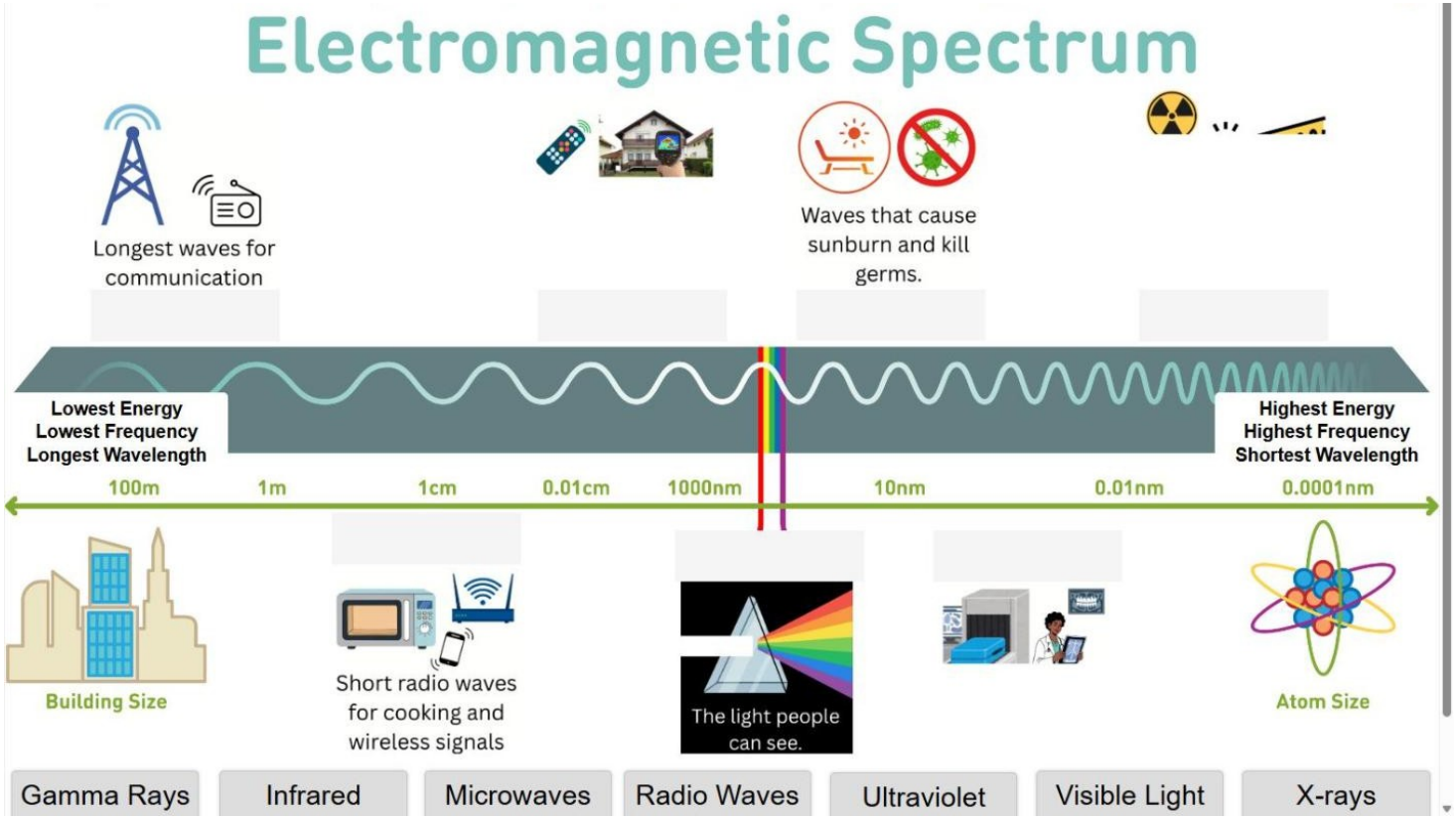
Single choice poll

C. gamma rays

Single choice poll

D. microwaves

Part 1: EM Spectrum Drag and Drop

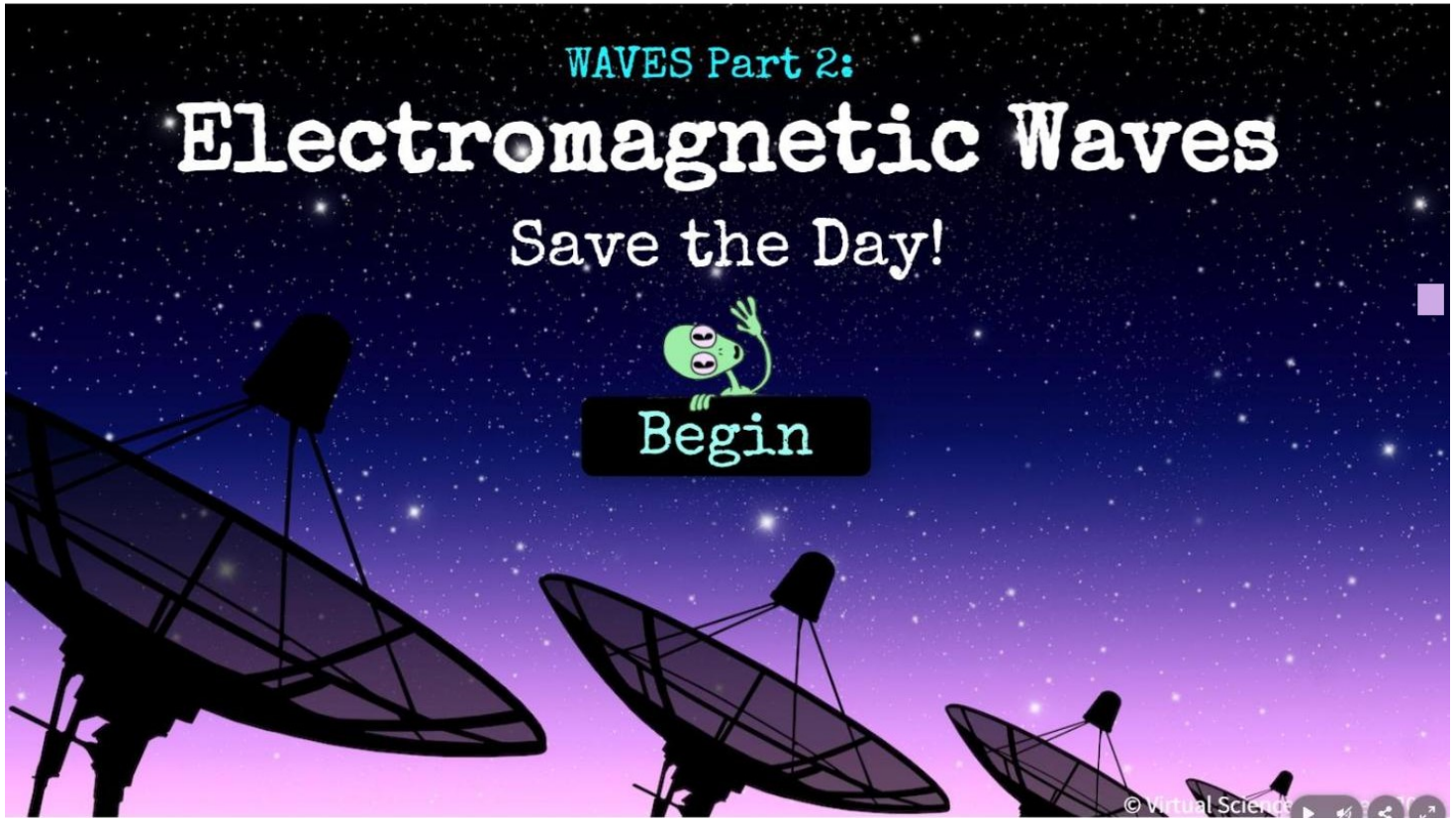


Open the EM Spectrum drag and drop

About 15 minutes

1. Drag each wave onto the spectrum. Press Check.
2. Q2: Use the pictures to match each clue to its wave.
3. Q3: Use the size scale (building to atom).

Part 2: EM Spectrum Escape Room



▶️ Open the EM Spectrum Escape Room

Use the rest of class

1. Click Begin. Use your notes and Part 1 to solve each puzzle.
2. Keep going until you escape.
3. The LAST screen is a certificate. Type your name.

Paste Your Certificate in Classkick

1. Snip it

Windows + Shift + S

Chromebook: Ctrl + Shift + Show windows. Drag a box around the certificate.

2. Copy it

Ctrl + C

If it did not copy already.

3. Paste it

Ctrl + V

Click the box on the Classkick slide, then paste. Your name must show.

Key Ideas

1. All EM waves travel through space at the speed of light. They need no medium.
2. Order: radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.
3. Higher frequency = shorter wavelength = more energy.
4. Radio waves can be as long as a building. Gamma rays are as small as an atom.

Re-rate your goals. Not done with the escape room? Finish it and paste your certificate.